

KEERTHIVASAN MURUGESAN

Chennai, India (Open to Relocate to Germany)

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Summary

Senior Data Engineer with 7+ years building batch ETL pipelines and curated datasets for risk, reporting, and analytics. Delivers Spark and PySpark transformations with SQL-first modeling, strong data quality gates, and stable batch operations. Experienced across HDFS and Hive, Parquet and Delta Lake, Databricks SQL Warehouse, and AWS migration patterns. Optimizes performance through partition strategy, caching, and join tuning to improve runtime and reliability. Partners with stakeholders to translate requirements into documented, consumption-ready tables.

Core Skills

- **Data Engineering:** ETL/ELT, batch processing, curated datasets, data lake, Parquet, Delta Lake
- **Data Quality:** Schema validation, reconciliation, null handling, consistency checks, publish gates
- **Cloud & Platforms:** AWS EMR, S3, Athena, Redshift, Databricks, BigQuery, HDFS, Hive
- **Orchestration & CI/CD:** Airflow, Jenkins, CI/CD pipelines, release hygiene, job stability
- **Programming:** Python, PySpark, Apache Spark, Spark SQL, SQL, pandas, NumPy, pytest
- **Delivery:** Agile execution, stakeholder management, requirements translation, documentation

Experience

IBM (Banking)

May 2024 – Present

Senior Data Engineer

Chennai, India

- Own batch risk analytics ETL pipelines on Spark, publishing curated datasets for reporting and modeling.
- Engineer PySpark transforms on HDFS and Hive tables, standardizing Parquet schemas and partition keys.
- Optimize joins and caching, cutting runtime by 25 to 40% and reducing batch failures by 15%.
- Support AWS migration to EMR and Athena, enabling self-serve queries for 10+ stakeholders.

Cognizant (Life Sciences & Healthcare)

January 2022 – May 2024

Senior Data Engineer

Chennai, India

- Built enterprise analytics pipelines delivering curated datasets for life sciences reporting and monitoring.
- Rebuilt legacy SQL and Informatica logic into maintainable MySQL transforms and Databricks Delta tables.
- Accelerated analytics iteration in SQL Warehouse, improving query turnaround by 25% for dashboards.
- Owned CI/CD for Databricks and AWS outputs, supporting weekly releases with controlled rollbacks.

Honeywell Technology Solutions (Aerospace)

April 2019 – January 2022

Senior Engineer

Chennai, India

- Delivered data pipelines and ML automation to support aerospace verification and predictive maintenance.
- Conducted testing and evaluation to verify the functionality and reliability of the MS&C system.
- Automated verification tooling, improving engineering productivity by 40% and reducing manual steps.
- Deployed PySpark anomaly models, improving detection lead time by hours to days for maintenance teams.

CADS Software Pvt Ltd

May 2018 – February 2019

Software Engineer

Chennai, India

- Built and supported Windows application modules and delivery automation for product releases.
- Developed .NET features and implemented Jenkins CI with Airflow scheduling for repeatable deployments.
- Stabilized releases through RCA-driven fixes and documentation, reducing recurring support tickets by 10%.

Projects

LOANS HRAM (High-Risk Account Management)

- Designed and deployed PySpark ETL on on-prem Hadoop for LOANS HRAM, processed 20M+ records/day on HDFS.
- Enabled real-time policy code calculations and proactive risk management for loans and Barclaycard portfolio.
- Optimized HDFS layout using Parquet + Snappy compression to cut storage and improve read performance.

Predictive Maintenance Data Product

- Built an end-to-end dataset and modeling pipeline for equipment anomaly detection.
- Engineered feature pipelines and data cleansing in Python and PySpark across multi-source logs.
- Operationalized anomaly signals for consumers, improving monitoring coverage by 20%.

Education

SRM University

June 2014 – April 2018

Bachelor of Engineering, Computer Science Engineering

Chennai, India

Certifications

- Databricks Data Engineering Certification, Databricks (2022)
- Data Science and AI Certification, IIT Madras via Intellipaat
- Python Programming, Coursera (2021)
- Lean Six Sigma Green Belt, Honeywell (2020)

Awards

- Peer recognition awards for execution and growth mindset

Languages

- English - Full Professional Proficiency (C2)
- German - Intermediate Working Proficiency (B1)